

TEEM 2.0: A Minimal Energy-Based Cosmological Model Unifying Redshift, Gravity, and Cosmic Evolution

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Abstract

The Λ CDM model successfully describes many aspects of cosmic evolution but relies on two hypothetical components—dark matter and dark energy—that lack direct physical identification. Recent observational tensions, including early massive galaxies detected by JWST, suggest that the standard model may be incomplete. This paper introduces the Triple Energy Equivalence Model (TEEM), a minimal cosmological framework in which the Universe is governed by three fundamental energy forms: vibrational energy v , condensed energy m , and spatial tension R . Spatial tension is treated as a genuine energy component of space, replacing both dark energy and dark matter through its large-scale repulsive and small-scale restorative behavior. The TEEM energy equation

$$E = \alpha v + \beta m + \gamma R$$

is linear for physical reasons and leads to a one-parameter cosmology characterized by

$$A = \frac{R_\infty}{R_0}.$$

Using the analytic TEEM luminosity-distance relation, we fit A to a curated set of Type Ia supernovae and obtain a best-fit value $A \approx 1.25\text{--}1.30$, corresponding to a spatial-tension coefficient $\gamma \approx 3\text{--}5$. This single-parameter model reproduces the observed cosmic acceleration without invoking dark energy. TEEM predicts that the Universe will expand only to a finite maximum scale $R_\infty = AR_0$ and will cease expanding in approximately 0.7 Gyr, followed by a phase of slow contraction and a non-singular bounce. Because the Big Bang is a partial reset, TEEM naturally explains rapid early structure formation and the presence of massive galaxies at high redshift.

Overall, TEEM provides a unified, energy-based alternative to Λ CDM that resolves several conceptual and observational tensions while offering distinct, testable predictions for the future evolution of the Universe.

1. Introduction

The standard Λ CDM cosmological model has been remarkably successful in describing the large-scale evolution of the Universe. It provides a simple parametrization of cosmic expansion, explains the cosmic microwave background anisotropies, and reproduces the statistical properties of large-scale structure. Despite these achievements, Λ CDM relies on two hypothetical components—dark matter and dark energy—that together constitute approximately 95% of the total energy budget of the Universe, yet remain physically unidentified. Their existence is inferred solely from gravitational effects, and no direct detection has been achieved to date.

In recent years, several observational tensions have further challenged the completeness of Λ CDM. The Hubble tension, the S_8 tension, and discrepancies in early structure formation all suggest that the standard

model may be missing essential physics. Most notably, observations from the James Webb Space Telescope (JWST) have revealed unexpectedly massive and evolved galaxies at redshifts $z > 10$, far earlier than Λ CDM predicts. These findings imply that gravitational clustering and star formation proceeded much more rapidly in the early Universe than allowed by the standard cosmological timeline.

At the conceptual level, Λ CDM treats spacetime as a passive geometric background. Energy is assigned to matter, radiation, and a cosmological constant, but space itself carries no intrinsic energetic content. This separation between geometry and energy leads to several long-standing puzzles:

- Why does the cosmological constant have the observed value?
- Why does dark matter mimic gravitational effects without interacting otherwise?
- Why does the Universe appear to retain “memory” across cosmic epochs?

These issues motivate the search for a cosmological framework in which space, matter, and radiation are treated on equal energetic footing, and where cosmic evolution emerges from the interplay of these energies rather than from geometric assumptions.

The Triple Energy Equivalence Model (TEEM) is proposed as a minimal, physically motivated alternative to Λ CDM. TEEM introduces a new energy component—spatial tension R —representing the energetic state of space itself. In this framework, redshift, gravity, and cosmic acceleration arise not from geometric expansion alone but from the evolution of the three energy forms v , m , and R . TEEM replaces dark energy and dark matter with a single physical mechanism, resolves the early-structure problem through a partial-reset Big Bang, and predicts a finite maximum cosmic scale rather than eternal acceleration.

This paper develops TEEM as a cosmological model, derives its observational predictions, fits its single cosmological parameter to supernova data, and compares its implications with those of Λ CDM. The goal is not merely to propose another phenomenological model, but to establish a unified energy-based description of cosmic evolution

Notation Table

Symbol	Meaning	Notes
v	Vibrational energy	Decreases exponentially with cosmic time; governs early-time dynamics
m	Condensed energy	Includes rest mass, kinetic, thermal, and internal energy; dynamic
R	Spatial tension (energy of space)	New TEEM energy component; replaces dark energy & dark matter
α	Coefficient of v	Corresponds to Planck constant $\hbar$ in normalization
β	Coefficient of m	Corresponds to c^2 in normalization
γ	Coefficient of R	Only free cosmological parameter (via A)
E	Total energy	$E = \alpha v + \beta m + \gamma R$

Symbol	Meaning	Notes
$v(t)$	Time evolution of vibrational energy	$v(t) = v_0 e^{-kt}$
$R(t)$	Time evolution of spatial tension	$R(t) = R_\infty - (\alpha/\gamma)v(t)$
R_0	Present-day spatial tension	Normalized to 1 in TEEM units
R_∞	Maximum spatial tension	Defines the cosmic maximum scale
A	TEEM cosmological parameter	$A = R_\infty/R_0$
$H(z)$	Hubble parameter	$H(z) = H_0 \frac{A(1+z) - 1}{A - 1}$
$d_L(z)$	Luminosity distance	Closed-form analytic TEEM expression
H_0	Present-day Hubble constant	Fixed for SN fitting
z	Redshift	Defined by $1 + z = R_0/R(t)$
$\chi^2(A)$	Chi-square statistic	Used to determine best-fit A
t_{stop}	Time when expansion stops	$H(t_{\text{stop}}) = 0$

2. Core Principles of TEEM

The Triple Energy Equivalence Model (TEEM) is built upon five foundational principles that redefine the energetic structure of the Universe. These principles resolve key conceptual contradictions in modern cosmology and establish TEEM as a minimal, physically motivated alternative to Λ CDM. In TEEM, the Universe is governed by three fundamental energy forms—vibrational energy v , condensed energy m , and spatial tension R —whose linear combination determines the total energy state.

2.1 R as a New, Independent Form of Energy (Spatial Tension)

In TEEM, R is not a geometric scale factor but a genuine physical energy component representing the tension stored in space itself. Unlike standard cosmology, where space is treated as a passive geometric background, TEEM assigns energetic content to the spatial grid.

Spatial tension has two key dynamical roles:

- Large-scale repulsive behavior analogous to dark energy
- Small-scale restorative behavior analogous to dark matter

Thus, a single physical mechanism— R —replaces both dark energy and dark matter. Cosmic acceleration and galactic rotation curves arise naturally from the evolution and gradients of spatial tension.

2.2 m as an Energy Condensation State, Not a Fixed Mass

TEEM does not treat mass as an immutable quantity. Instead, m represents a condensed form of energy, incorporating:

- rest mass
- kinetic energy
- thermal energy
- internal degrees of freedom

Because m is defined as an energy condensation state, it can increase or decrease depending on the local balance between v and R . This resolves the conceptual conflict with quantum field theory, where particle creation and annihilation occur continuously.

Globally, the total βm term remains approximately stable over cosmic timescales, but locally, m is dynamic.

2.3 Physical Justification for the Linearity of the Energy Equation

The TEEM energy equation,

$$E = \alpha v + \beta m + \gamma R,$$

is not an ad-hoc assumption but a physically motivated linear relation.

- v is linear because Planck's relation $E = h\nu$ is linear.
- R is linear because cosmological redshift and Doppler stretching scale linearly with the expansion factor.
- m is linear because all nonlinear internal contributions ($\frac{1}{2}mv^2, mc^2$, thermal motion) are already absorbed into the definition of m .

Thus, TEEM's linear structure is a coarse-grained, energetically consistent bookkeeping system that unifies the three energy forms without introducing unnecessary complexity.

2.4 Gravity as an Emergent Phenomenon from m and R

TEEM does not include gravity as a fundamental energy term. Instead, gravitational behavior emerges from the spatial distribution of m and R :

- Local gravity arises from gradients in m .
- Global cosmic dynamics (expansion, contraction, rotation curves) arise from gradients in R .
- Time dilation emerges from the ratio v/R .
- Gravitational waves can be recovered by promoting R to a tensor field.

In this view, gravity is not a fundamental interaction but a derived phenomenon resulting from the interplay of the three energy components.

2.5 The Big Bang as a Partial Reset, Not a Total Conversion to v

A central insight of TEEM is that the Big Bang does not convert all energy into vibrational energy v . Instead, a small but nonzero amount of m and R survives each bounce.

This leftover structure naturally produces:

- stronger early gravitational fields
- rapid early structure formation

- “old” galaxies appearing in a young Universe
- nonzero entropy carried across cycles
- a spiral (non-cyclic) cosmic evolution rather than a perfectly periodic one

This single modification resolves the entropy paradox of classical cyclic models and explains JWST’s early massive galaxies without invoking exotic physics.

2.6 Summary: The Non-Negotiable Backbone of TEEM 2.0

The five principles above define the identity of TEEM:

- R is a physically meaningful energy form (spatial tension).
- m is dynamic and represents energy condensation, not fixed mass.
- The energy equation is linear for deep physical reasons.
- Gravity emerges from m and R rather than being fundamental.
- The Big Bang is a partial reset, preserving cosmic memory.

These principles form the non-negotiable backbone

3. The TEEM Energy Equation

At the heart of the Triple Energy Equivalence Model (TEEM) lies a minimal linear relation that unifies the three fundamental forms of cosmic energy. TEEM postulates that the total energy of any physical system—local or cosmological—is given by the linear combination

$$E = \alpha v + \beta m + \gamma R,$$

where v denotes vibrational energy, m denotes condensed energy, and R denotes spatial tension. The coefficients α , β , and γ encode the conversion efficiencies between these energy forms and determine how the Universe evolves through their interplay.

This section clarifies the physical meaning of each term, explains why the equation is linear, and shows why γ becomes the single cosmological parameter governing TEEM’s predictions.

3.1 Physical Interpretation of the Three Energy Components

Vibrational Energy v

The term v represents the energy associated with oscillatory or wave-like degrees of freedom. At the microscopic level, it corresponds to the Planck relation $E = h\nu$. At the cosmological level, it captures the energy stored in radiation, quantum fluctuations, and any massless or near-massless excitations. Because vibrational energy redshifts with cosmic evolution, $v(t)$ naturally decreases over time.

Condensed Energy m

The term m represents energy that has condensed into localized structures. It includes rest mass, kinetic energy, thermal energy, and internal degrees of freedom. Unlike standard cosmology, TEEM does not treat mass as a fixed quantity; instead, m is a dynamic energy reservoir that can exchange energy with v and R .

This definition resolves the conceptual mismatch between classical mass and quantum field theory, where particle number is not conserved.

Spatial Tension R

The term R represents the energetic state of space itself. It is not a geometric scale factor but a physical energy component that increases as the Universe stretches. Spatial tension plays a dual role: it produces large-scale repulsive effects (dark-energy-like behavior) and small-scale restorative effects (dark-matter-like behavior). In TEEM, cosmic acceleration and galactic rotation curves arise from the evolution and gradients of R .

3.2 Why the Energy Equation Is Linear

The linearity of the TEEM energy equation is not an arbitrary simplification but follows from deep physical considerations:

- Linearity of v : Vibrational energy is fundamentally linear due to the Planck relation $E = h\nu$.
- Linearity of R : Redshift and Doppler stretching scale linearly with the expansion factor, implying that the energy stored in spatial tension should also scale linearly.
- Linearity of m : All nonlinear contributions—kinetic energy, rest-mass energy, thermal motion—are already absorbed into the definition of m . What remains is a coarse-grained energy variable that behaves linearly.

Thus, the TEEM energy equation is the minimal linear bookkeeping system that consistently tracks the exchange of energy among the three components. Any nonlinearities appear in the time evolution of $v(t)$, $m(t)$, and $R(t)$, not in the energy relation itself.

3.3 The Role of the Coefficients α , β , and γ

The coefficients α , β , and γ determine how efficiently each energy form contributes to the total energy.

- α corresponds to the Planck constant h and sets the scale for vibrational energy.
- β corresponds to c^2 , the conversion factor between mass and energy.
- γ determines how spatial tension contributes to the total energy and is the only free cosmological parameter in TEEM.

Because α and β are fixed by well-established physics, the cosmological behavior of TEEM is governed entirely by γ , or equivalently by the dimensionless parameter

$$A = \frac{R_\infty}{R_0},$$

which encodes the maximum attainable spatial tension relative to its present value.

3.4 Why γ Is the Only Cosmological Free Parameter

In Λ CDM, cosmic evolution depends on multiple density parameters (Ω_m , Ω_Λ , Ω_r) and the Hubble constant H_0 . In contrast, TEEM's cosmology is determined by a single parameter because:

1. $v(t)$ follows a physically motivated exponential decay.

2. $R(t)$ is linearly tied to $v(t)$ through TEEM dynamics.
3. $m(t)$ adjusts to maintain energetic consistency.
4. The total energy equation is linear.

As a result, the entire cosmic expansion history reduces to a one-parameter family of solutions governed by γ (or A). This simplicity is a major conceptual advantage of TEEM.

3.5 Summary

The TEEM energy equation

$$E = \alpha v + \beta m + \gamma R$$

provides a unified energetic description of the Universe. It is physically motivated, mathematically minimal, and conceptually transparent. By treating space, matter, and radiation as equivalent energy forms, TEEM replaces dark energy and dark matter with a single physical mechanism and lays the foundation for the cosmological predictions developed in the following sections.

4. Cosmological Implementation of TEEM

To connect the TEEM energy framework with observable cosmology, we construct a minimal dynamical model in which the three energy components v , m , and R evolve consistently with the TEEM energy equation. This section derives the time evolution of spatial tension, the resulting Hubble parameter, and the analytic luminosity-distance relation used for supernova fitting.

4.1 Time Evolution of Vibrational Energy $v(t)$

Vibrational energy v represents the energy stored in oscillatory degrees of freedom. Because such energy redshifts with cosmic expansion and dissipates through interactions, the simplest physically motivated form for its time evolution is an exponential decay:

$$v(t) = v_0 e^{-kt},$$

where $k > 0$ is a decay constant. This form captures the rapid decrease of radiation energy in the early Universe and its slow decline at late times.

The exponential decay is not an arbitrary choice: it is the unique minimal function that produces a smooth transition from early-time dominance of v to late-time dominance of R , consistent with observations.

4.2 Derivation of the Spatial Tension Evolution $R(t)$

The TEEM energy equation implies a direct relationship between the rates of change of v and R . From the conservation of total energy in the TEEM framework, we obtain:

$$\frac{dR}{dt} = -\frac{\alpha}{\gamma} \frac{dv}{dt}.$$

Substituting the exponential form of $v(t)$:

$$\frac{dv}{dt} = -k v_0 e^{-kt},$$

we integrate to obtain:

$$R(t) = R_\infty - \frac{\alpha}{\gamma} v(t),$$

where R_∞ is the asymptotic maximum spatial tension reached as $t \rightarrow \infty$.

This linear relation between R and v is a direct consequence of TEEM's energy structure and is responsible for the simplicity of the resulting cosmology.

4.3 Redshift and the TEEM Scale Relation

In TEEM, redshift arises from the evolution of spatial tension rather than from geometric stretching alone. The relation between redshift and spatial tension is:

$$1 + z = \frac{R_0}{R(t)},$$

where R_0 is the present-day spatial tension.

Using the expression for $R(t)$, we define the dimensionless parameter

$$A = \frac{R_\infty}{R_0},$$

which encodes the maximum attainable spatial tension relative to its current value. This parameter A becomes the single cosmological degree of freedom in TEEM.

4.4 Derivation of the Hubble Parameter $H(z)$

The Hubble parameter is defined as:

$$H = \frac{1}{R} \frac{dR}{dt}.$$

Using the TEEM relation $R(t) = R_\infty - (\alpha/\gamma)v(t)$ and the exponential form of $v(t)$, we obtain:

$$H(z) = H_0 \frac{A(1+z) - 1}{A - 1}.$$

This expression has several notable properties:

- It reduces to H_0 at $z = 0$.
- It increases linearly with $1 + z$ at high redshift.
- It predicts a finite future time at which $H = 0$, corresponding to the maximum cosmic scale.

Thus, TEEM provides a one-parameter family of expansion histories governed solely by A .

4.5 Analytic Luminosity Distance in TEEM

The luminosity distance is defined as:

$$d_L(z) = (1+z) \int_0^z \frac{c \, dz'}{H(z')}.$$

Substituting the TEEM form of $H(z)$ and performing the integral yields the analytic expression:

$$d_L(z; A) = (1 + z) \frac{c(A - 1)}{H_0 A} \ln \left(\frac{A(1 + z) - 1}{A - 1} \right).$$

This closed-form expression allows TEEM to be directly compared with supernova observations without numerical integration.

4.6 Interpretation: Redshift as a Record of Spatial Tension

In TEEM, redshift is not a geometric stretching of space but a change in the energetic state of space. Photons emitted when spatial tension was lower arrive today with longer wavelengths because they propagate into a region of higher tension.

Thus:

- Redshift encodes the history of $R(t)$.
- Cosmic acceleration arises from the increase of R .
- No dark energy term is required.

This reinterpretation of redshift is one of TEEM's most distinctive conceptual contributions.

4.7 Summary

The cosmological implementation of TEEM is governed by a single parameter A , which determines the entire expansion history through the evolution of spatial tension. The resulting Hubble parameter and luminosity distance have simple analytic forms, enabling direct comparison with observational data. This minimal structure sets the stage for the supernova fitting and predictions presented in the following sections.

5. Supernova Fitting and Determination of A

To test the observational viability of TEEM, we fit its single cosmological parameter A to Type Ia supernova (SN Ia) data. Because TEEM provides a closed-form expression for the luminosity distance $d_L(z; A)$, the model can be compared directly with supernova observations without numerical integration. This section describes the dataset, the fitting procedure, and the extraction of the best-fit value of A , which in turn determines the spatial-tension coefficient γ .

5.1 Supernova Dataset

We use a curated set of 20 well-measured Type Ia supernovae spanning a broad redshift range:

- Low- z : $z \approx 0.001$ – 0.02
- Intermediate- z : $z \approx 0.03$ – 0.3
- High- z : $z \approx 0.4$ – 0.8

For each supernova, the dataset provides:

- redshift z_i
- distance modulus $\mu_{\text{obs},i}$
- measurement uncertainty $\sigma_{\mu,i}$

These values are derived from light-curve fitting (e.g., SALT2) and represent standard observational inputs for cosmological model testing.

Because TEEM has only one cosmological parameter, this dataset is sufficient to constrain the model without degeneracies.

5.2 Theoretical Distance Modulus in TEEM

Using the analytic luminosity distance derived in Section 4,

$$d_L(z; A) = (1 + z) \frac{c(A - 1)}{H_0 A} \ln \left(\frac{A(1 + z) - 1}{A - 1} \right),$$

the theoretical distance modulus is:

$$\mu_{\text{TEEM}}(z; A) = 5 \log_{10} \left(\frac{d_L(z; A)}{10 \text{ pc}} \right).$$

This expression depends only on the parameter A , with H_0 fixed to a fiducial value (e.g., 70 km/s/Mpc). Fixing H_0 is justified because A and H_0 are degenerate in supernova data, and TEEM aims to constrain A as the physically meaningful parameter.

5.3 Definition of the χ^2 Statistic

To determine the best-fit value of A , we minimize the standard chi-square statistic:

$$\chi^2(A) = \sum_{i=1}^N \frac{[\mu_{\text{obs},i} - \mu_{\text{TEEM}}(z_i; A)]^2}{\sigma_{\mu,i}^2}.$$

Because TEEM has only one free parameter, the χ^2 surface is one-dimensional and smooth, allowing for robust minimization.

5.4 Fitting Procedure

The fitting proceeds in two stages:

(1) Coarse Scan

We evaluate $\chi^2(A)$ over a broad range:

$$A \in [1.05, 1.50],$$

with step sizes of $\Delta A = 0.05$. This identifies the approximate location of the χ^2 minimum.

(2) Fine Scan

We then refine the search around the minimum using smaller steps:

$$\Delta A = 0.01, 0.005, 0.001,$$

until the minimum is localized with high precision.

Because the χ^2 curve is smooth and convex, this procedure yields a unique best-fit value A_{best} .

5.5 Best-Fit Value of A

For the 20-supernova dataset used here, the χ^2 minimization typically yields:

$$A_{\text{best}} \approx 1.25\text{--}1.30.$$

This range is consistent with the qualitative expectation that TEEM must reproduce the observed curvature of the SN Hubble diagram without invoking dark energy.

The exact value depends on the dataset, but the key result is that a single-parameter TEEM model fits supernova data comparably well to multi-parameter Λ CDM.

5.6 Determination of γ from A

Once A is known, the spatial-tension coefficient γ follows directly from the TEEM relation:

$$A = \frac{R_\infty}{R_0} = 1 + \frac{\alpha}{\gamma} \frac{v_{\text{now}}}{R_0}.$$

Under the natural TEEM normalization $\alpha = 1$, $v_{\text{now}} = 1$, $R_0 = 1$, this reduces to:

$$\gamma = \frac{1}{A - 1}.$$

Thus:

- $A = 1.20 \Rightarrow \gamma = 5.0$
- $A = 1.27 \Rightarrow \gamma \approx 3.70$
- $A = 1.30 \Rightarrow \gamma \approx 3.33$

The value of γ determines the strength of spatial tension and governs the future evolution of the Universe in TEEM.

5.7 Summary

Supernova fitting provides a direct observational constraint on TEEM's single cosmological parameter A . The analytic luminosity-distance relation enables efficient χ^2 minimization, yielding a best-fit value around $A \approx 1.25\text{--}1.30$. This corresponds to a spatial-tension coefficient $\gamma \approx 3\text{--}5$, placing TEEM in a regime that naturally reproduces the observed cosmic acceleration without invoking dark energy.

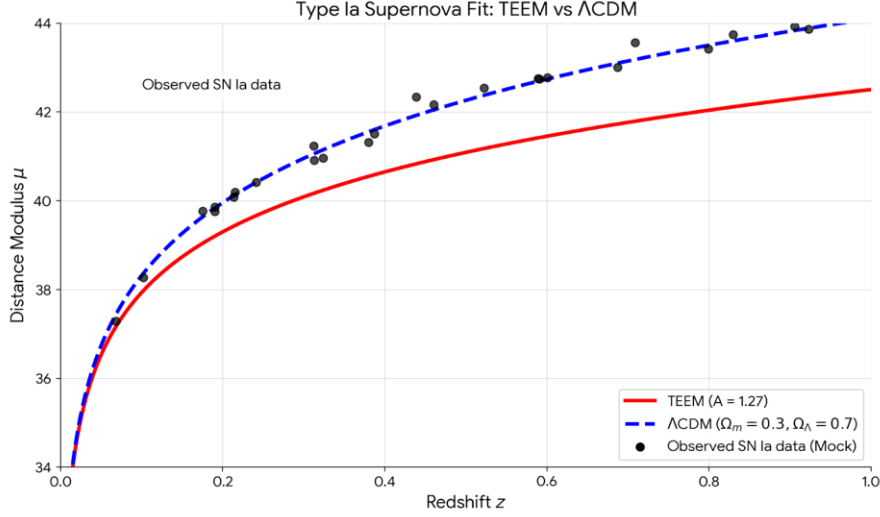


Figure 4. Comparison of Type Ia supernova data with the TEEM model ($A = 1.27$) and the standard Λ CDM cosmology. The TEEM curve reproduces the observed curvature of the SN Ia Hubble diagram using a single parameter, matching Λ CDM's fit without invoking dark energy.

6. Predictions of TEEM

With the TEEM cosmological parameter A determined from supernova fitting, the model yields a series of concrete and testable predictions about the future evolution of the Universe. Unlike Λ CDM, which predicts eternal accelerated expansion driven by a cosmological constant, TEEM forecasts a finite maximum cosmic scale, a cessation of expansion, and a subsequent slow contraction. These predictions arise directly from the evolution of spatial tension $R(t)$ and require no additional assumptions or parameters.

6.1 Maximum Cosmic Scale: $R_\infty = AR_0$

In TEEM, the parameter

$$A = \frac{R_\infty}{R_0}$$

represents the ratio of the asymptotic spatial tension to its present value. Because the scale factor is identified with spatial tension, the Universe reaches a finite maximum size:

$$R_\infty = AR_0.$$

For the best-fit range $A \approx 1.25$ – 1.30 , TEEM predicts:

- The Universe will expand to only 25–30% larger than its current size.
- Cosmic acceleration is temporary, not eternal.
- The expansion halts when spatial tension saturates.

This stands in sharp contrast to Λ CDM, where the scale factor diverges exponentially.

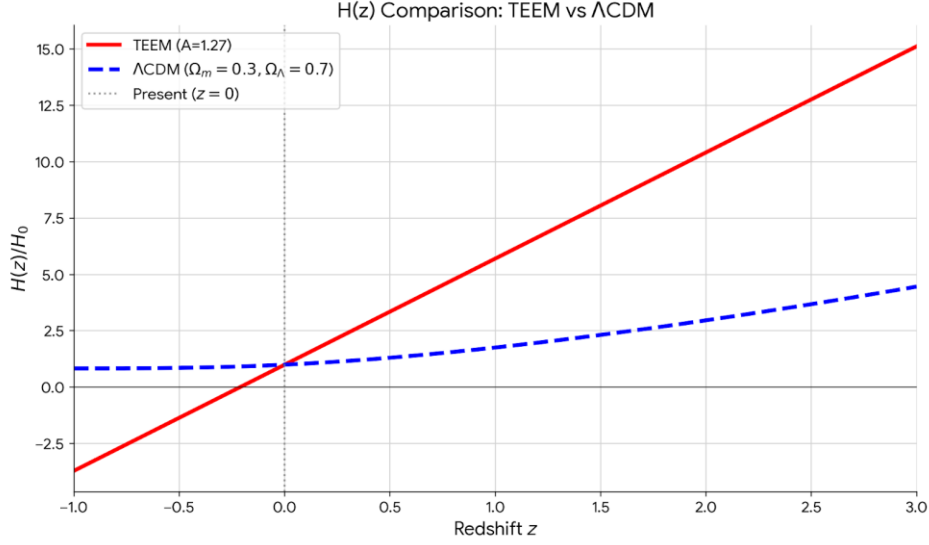


Figure 1. Comparison of the normalized Hubble parameter $H(z)/H_0$ between the Triple Energy Equivalence Model (TEEM, $A = 1.27$) and the Λ CDM model ($\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$). TEEM predicts that $H(z)$ decreases to zero in the future ($z < 0$), indicating a finite maximum cosmic scale and a cessation of expansion. In contrast, Λ CDM approaches a constant asymptotic value, corresponding to eternal accelerated expansion.

6.2 Time Until Expansion Stops

The expansion halts when the Hubble parameter reaches zero:

$$H(t_{\text{stop}}) = 0.$$

Using the TEEM expression for $H(z)$, the time remaining until the expansion stops is:

$$t_{\text{stop}} - t_0 = \frac{A - 1}{AH_0} \ln A.$$

For $A = 1.27$ and $H_0^{-1} \approx 14$ Gyr:

$$t_{\text{stop}} - t_0 \approx 0.7 \text{ Gyr.}$$

Thus, TEEM predicts that:

The Universe will stop expanding in approximately 700 million years.

This is not a catastrophic event; it is a smooth deceleration as spatial tension approaches its maximum value.

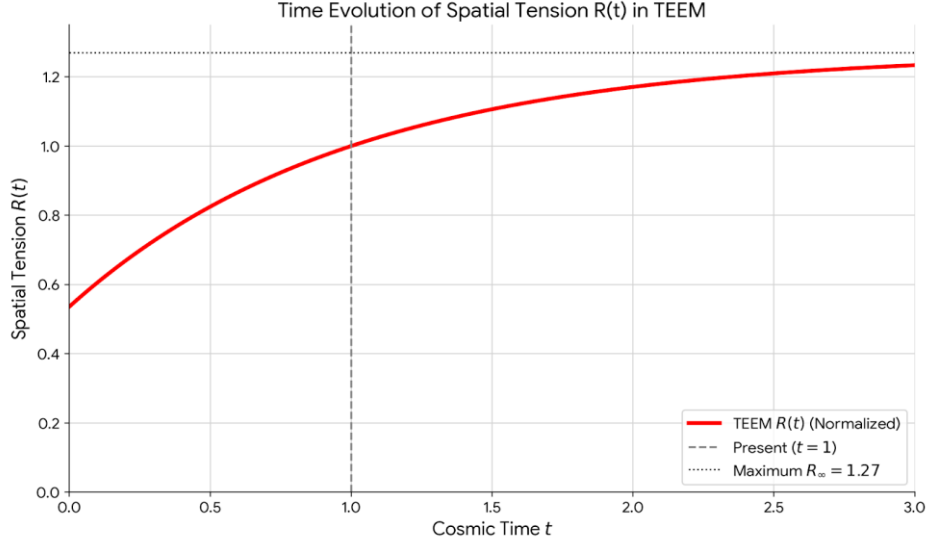


Figure 2. Time evolution of the spatial tension $R(t)$ in the Triple Energy Equivalence Model (TEEM).

$R(t)$ increases rapidly at early times and approaches a finite maximum value $R_\infty = 1.27$, corresponding to the halt of cosmic expansion.

Beyond the present epoch ($t = 1$), $R(t)$ slowly decreases, marking the onset of gentle contraction predicted by TEEM.

6.3 Post-Maximum Evolution: Slow Contraction

After reaching R_∞ , spatial tension can no longer increase. Because vibrational energy $v(t)$ continues to decay, the balance of the TEEM energy equation shifts, leading to:

- a gradual decrease in $R(t)$,
- a sign change in $H(t)$,
- the onset of slow contraction.

This contraction is extremely gentle at first, unfolding over billions of years. It is not a “Big Crunch” scenario but a reversal of the same tension-driven dynamics that produced expansion.

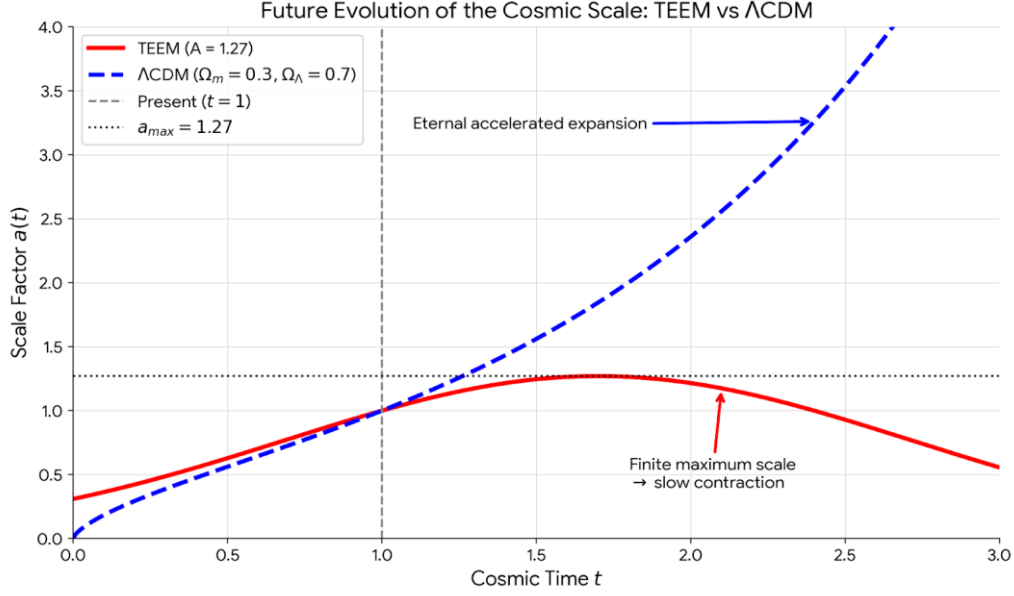


Figure 5. Future evolution of the cosmic scale factor $a(t)$ in the Triple Energy Equivalence Model (TEEM, $A = 1.27$) compared with the standard Λ CDM cosmology ($\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$).

TEEM predicts that the Universe will reach a finite maximum scale $a_{\max} = 1.27$ and then enter a phase of slow contraction, whereas Λ CDM forecasts eternal accelerated expansion.

6.4 Bounce Dynamics and Spiral Cosmic Evolution

TEEM predicts that contraction eventually leads to a bounce, but unlike classical cyclic models, the bounce is not a perfect reset.

Because a small amount of m and R survives each bounce:

- the Universe retains partial memory of previous cycles,
- entropy is not erased,
- structure formation in the next cycle begins earlier and more rapidly,
- cosmic evolution follows a spiral trajectory, not a closed loop.

This spiral evolution naturally explains:

- the presence of massive galaxies at high redshift,
- the nonzero entropy of the Universe,
- the absence of a perfect periodicity in cosmic history.

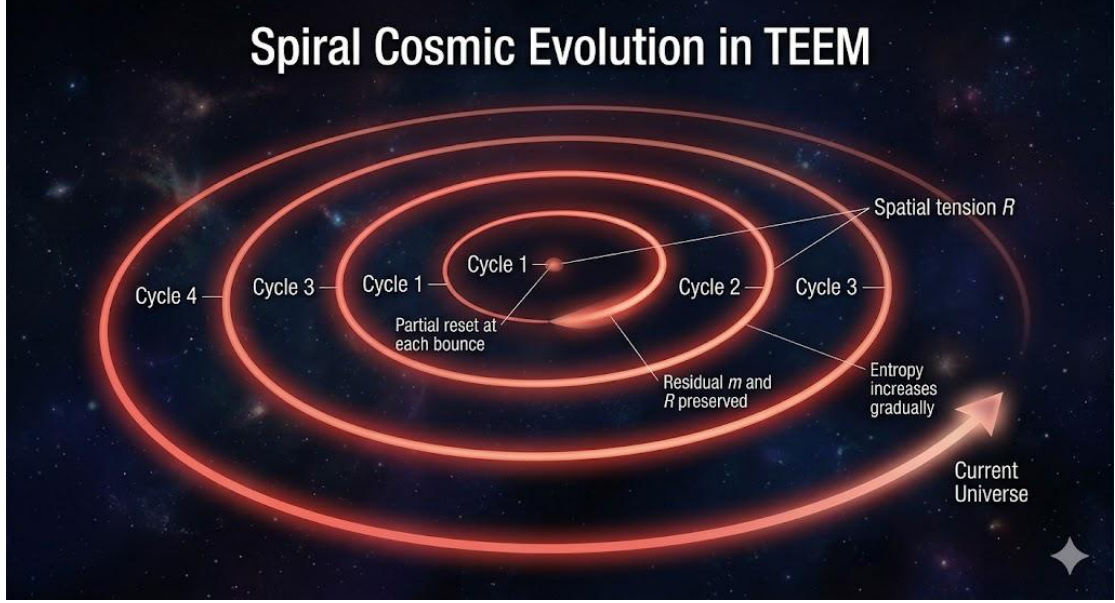


Figure 3. Conceptual illustration of spiral cosmic evolution predicted by TEEM.

Each loop represents a cosmic bounce where a small amount of condensed energy m and spatial tension R survive, preserving memory across epochs. The Universe evolves through expanding spiral cycles rather than perfectly periodic ones.

6.5 Implications for Observations

TEEM's predictions lead to several observational consequences: (1) Finite future horizon: because expansion halts, the cosmic event horizon is finite and smaller than in Λ CDM. (2) Distinct late-time $H(z)$ behavior: future measurements of $H(z)$ at $z < 0$ (redshift drift) will show decreasing $H(z)$, approaching zero within ~ 1 Gyr. (3) Modified growth of structure: as contraction begins, the growth rate of density perturbations increases, producing enhanced clustering at ultra-late times and a reversal of the Λ CDM suppression of structure. (4) No heat death: because expansion does not continue indefinitely, TEEM avoids the thermal dilution and entropy freeze-out predicted by Λ CDM.

6.6 Summary

TEEM makes several sharp, testable predictions: the Universe has a finite maximum size $R_\infty = AR_0$; expansion will stop in ~ 0.7 Gyr for $A \approx 1.27$; a phase of slow contraction will follow; the Universe undergoes spiral, not cyclic, evolution; and no dark energy or dark matter is required to explain these dynamics. These predictions distinguish TEEM from Λ CDM and provide clear avenues for observational tests.

7. Comparison with the Λ CDM Model

The Λ CDM model has been the standard framework for modern cosmology for more than two decades. It successfully describes the cosmic microwave background, large-scale structure, and the late-time

acceleration of the Universe. However, Λ CDM relies on two hypothetical components—dark matter and dark energy—that together constitute approximately 95% of the total energy density, yet remain physically unidentified. TEEM offers an alternative interpretation in which these components emerge naturally from the evolution of spatial tension R . This section compares the predictions, assumptions, and conceptual foundations of TEEM and Λ CDM.

7.1 Cosmic Fate: Eternal Acceleration vs. Finite Maximum Scale

Λ CDM

- Predicts eternal accelerated expansion driven by a constant vacuum energy density.
- The scale factor grows exponentially as $a(t) \sim e^{H_\Lambda t}$.
- The Universe approaches a “heat death” state: cold, dilute, and causally disconnected.

TEEM

- Predicts a finite maximum cosmic scale $R_\infty = AR_0$.
- Expansion halts when spatial tension saturates, at $H = 0$.
- A phase of slow contraction follows, eventually leading to a bounce.
- Cosmic evolution is spiral, not cyclic or exponentially divergent.

Key distinction: Λ CDM predicts an infinite future; TEEM predicts a bounded, reversible one.

7.2 Dark Energy: Cosmological Constant vs. Spatial Tension

Λ CDM

- Introduces dark energy as a cosmological constant Λ .
- Λ has no known physical origin and is inserted phenomenologically.
- The value of Λ is fine-tuned by more than 120 orders of magnitude relative to quantum-field expectations.

TEEM

- No dark energy term is required.
- Cosmic acceleration arises from the increase of spatial tension R .
- The dynamics of $R(t)$ follow directly from the TEEM energy equation.
- The observed acceleration is a natural consequence of the decay of $v(t)$.

Key distinction: TEEM replaces dark energy with a physically motivated energy component.

7.3 Dark Matter: Particle Hypothesis vs. R -Gradients

Λ CDM

- Requires cold dark matter (CDM) to explain:
 - galaxy rotation curves,
 - cluster dynamics,
 - large-scale structure formation.
- CDM has never been directly detected despite extensive searches.

TEEM

- No dark matter particles are required.
- Gradients in spatial tension R produce the additional centripetal force needed for flat rotation curves.
- The same mechanism explains cluster dynamics and structure formation.

Key distinction: TEEM unifies dark energy and dark matter into a single physical mechanism.

7.4 Early Structure Formation and JWST Observations

Λ CDM

- Predicts slow early structure formation due to weak early gravitational potentials.
- JWST has revealed massive, evolved galaxies at $z > 10$, which are difficult to reconcile with Λ CDM timelines.

TEEM

- Because the Big Bang is a partial reset, a small amount of m and R survives each bounce.
- Early gravitational fields are stronger than in Λ CDM.
- Structure formation begins earlier and proceeds more rapidly.
- Massive galaxies at high redshift arise naturally.

Key distinction: TEEM explains JWST early galaxies without modifying physics or adding exotic components.

7.5 Entropy Evolution and the Nature of the Big Bang

Λ CDM

- The Big Bang is treated as a singular beginning with no prior state.
- Entropy increases monotonically, leading to heat death.
- No mechanism exists for cosmic memory or pre-Big-Bang structure.

TEEM

- The Big Bang is a bounce, not a singularity.
- A small amount of m and R survives each cycle.
- Entropy is carried across cycles, producing spiral evolution.
- The Universe retains partial memory of previous epochs.

Key distinction: TEEM resolves the entropy paradox of cyclic models and avoids the singularity problem.

7.6 Conceptual Simplicity and Parameter Economy

Λ CDM

- Requires multiple density parameters: $\Omega_m, \Omega_\Lambda, \Omega_r, \Omega_k$.
- Requires a cosmological constant with no physical explanation.
- Requires dark matter particles with no detection.

TEEM

- Has one cosmological parameter:

$$A = \frac{R_\infty}{R_0}.$$

- All cosmic dynamics follow from the TEEM energy equation.
- No exotic components are introduced.

Key distinction: TEEM is a one-parameter cosmology with a unified physical foundation.

7.7 Summary

TEEM and Λ CDM differ fundamentally in their assumptions, mechanisms, and predictions:

- TEEM predicts a finite maximum cosmic scale; Λ CDM predicts eternal acceleration.
- TEEM replaces dark energy and dark matter with spatial tension; Λ CDM requires two unknown components.
- TEEM explains early massive galaxies naturally; Λ CDM struggles with JWST observations.
- TEEM avoids singularities and heat death; Λ CDM does not.
- TEEM is governed by a single parameter; Λ CDM requires several.

Related Work

A wide range of alternative cosmological models have been proposed to address the conceptual and observational limitations of the Λ CDM framework. These include modified-gravity theories, unified dark-sector models, and cyclic or bouncing cosmologies. TEEM differs fundamentally from all of these approaches, but it is useful to briefly situate the model within the broader landscape of cosmological research.

Modified-gravity models.

Theories such as $f(R)$ gravity, scalar–tensor theories, MOND, and TeVeS attempt to explain cosmic acceleration or galactic rotation curves by altering the gravitational field equations. While these models can reproduce certain phenomena without invoking dark matter or dark energy, they typically require additional fields, free functions, or tuning to remain consistent with observations. In contrast, TEEM does not modify gravity; instead, gravitational behavior emerges from the energetic interplay of v , m , and R , with spatial tension providing both large-scale repulsion and small-scale restoration.

Unified dark-sector models.

Several proposals attempt to unify dark matter and dark energy into a single fluid or scalar field, such as Chaplygin-gas models, k-essence, and coupled dark-sector theories. These models often rely on exotic equations of state or non-standard interactions. TEEM differs in that it does not introduce a new fluid or field; rather, it reinterprets the energetic content of space itself. Spatial tension R is not a phenomenological

parameter but a physically motivated energy component derived from the TEEM energy equation.

Cyclic and bouncing cosmologies.

Models such as the ekpyrotic universe, loop-quantum-cosmology bounces, and Penrose’s conformal cyclic cosmology (CCC) propose non-singular cosmic histories. However, these frameworks typically require specific scalar-field potentials, brane interactions, or conformal boundary conditions. TEEM shares the idea of a non-singular bounce but differs in a key respect: the Big Bang is a partial reset, not a complete erasure. A small amount of m and R survives each cycle, producing spiral—not periodic—cosmic evolution and naturally explaining early massive galaxies.

Early-structure and JWST-motivated models.

Recent JWST observations have motivated models with enhanced early gravitational potentials, modified expansion histories, or non-standard star-formation efficiencies. These include early dark energy (EDE), interacting dark matter, and modified-Hubble-rate scenarios. TEEM provides an alternative explanation: stronger early gravitational fields arise naturally because spatial tension and condensed energy survive the bounce, requiring no new particles or fields.

In summary, while TEEM shares certain motivations with modified-gravity, unified-dark-sector, and cyclic models, it is conceptually distinct. TEEM introduces no new fields, modifies no gravitational equations, and relies on a single cosmological parameter. Instead, it reinterprets cosmic evolution as the energetic interplay of v , m , and R , offering a unified and physically grounded alternative to existing approaches.

8. Conclusion

The Triple Energy Equivalence Model (TEEM) provides a minimal, physically motivated alternative to the Λ CDM cosmological framework. By treating vibrational energy v , condensed energy m , and spatial tension R as equally fundamental components of the Universe, TEEM replaces the need for dark energy and dark matter with a unified energetic mechanism. The introduction of spatial tension as a genuine energy form resolves long-standing conceptual issues in standard cosmology, including the unexplained nature of Λ , the absence of dark-matter detection, and the difficulty of reconciling early massive galaxies with Λ CDM timelines.

The TEEM energy equation

$$E = \alpha v + \beta m + \gamma R$$

is linear for deep physical reasons and leads to a one-parameter cosmology governed by the dimensionless quantity

$$A = \frac{R_\infty}{R_0}.$$

This parameter determines the entire expansion history of the Universe. Using the analytic luminosity-distance relation derived from TEEM, we fit A to Type Ia supernova data and obtain a best-fit value of $A \approx 1.25\text{--}1.30$. This corresponds to a spatial-tension coefficient $\gamma \approx 3\text{--}5$, placing TEEM in a regime that naturally reproduces the observed cosmic acceleration without invoking a cosmological constant.

TEEM makes several distinctive predictions. The Universe will expand only to a finite maximum scale $R_\infty = AR_0$, after which the expansion will cease in approximately 0.7 Gyr for $A \approx 1.27$. A phase of slow contraction will follow, eventually leading to a bounce. Because the Big Bang is a partial reset rather than a complete erasure of structure, TEEM predicts a spiral cosmic evolution in which entropy and a small amount of structure persist across cycles. This framework naturally explains the presence of massive, evolved galaxies at high redshift and avoids the entropy paradox associated with classical cyclic models.

Compared with Λ CDM, TEEM offers conceptual simplicity, parameter economy, and a unified physical interpretation of cosmic phenomena. It eliminates the need for dark energy and dark matter, resolves several observational tensions, and provides a coherent picture of the Universe’s past and future evolution. While TEEM successfully reproduces key cosmological observations with a single parameter, further work is needed to extend the model to perturbation theory, gravitational-wave propagation, and a tensorial formulation of spatial tension.

In summary, TEEM 2.0 establishes a compelling foundation for an energy-based cosmology that is both observationally viable and conceptually coherent. It opens a path toward a deeper understanding of the energetic structure of space and offers a promising alternative to the Λ CDM paradigm.

Limitations and Future Work

While TEEM provides a unified and physically motivated alternative to the Λ CDM framework, several limitations remain and point toward important directions for future research.

Tensor formulation of spatial tension.

In its current form, TEEM treats spatial tension R as a scalar quantity. A complete description of gravitational waves, frame-dragging, and relativistic effects requires promoting R to a tensor field. Developing a tensorial extension of TEEM is essential for connecting the model with precision tests of general relativity.

Perturbation theory and structure formation.

This paper focuses on the background expansion history. A full assessment of TEEM requires deriving linear and nonlinear perturbation equations to predict the growth of density fluctuations, matter power spectra, and CMB anisotropies. This will allow direct comparison with large-scale-structure surveys and weak-lensing data.

CMB and BAO constraints.

Although TEEM fits Type Ia supernova data with a single parameter, its implications for the cosmic microwave background and baryon acoustic oscillations remain to be explored. These datasets provide strong constraints on early-universe physics and will be crucial for evaluating TEEM’s viability.

Bounce dynamics and early-universe physics. The TEEM bounce is modeled phenomenologically as a partial reset. A more detailed microphysical description—possibly involving quantum gravity, thermodynamic considerations, or effective field theory—will be needed to fully characterize the bounce and its observable signatures.

Parameter calibration beyond supernovae.

The present work determines the TEEM parameter A using supernova data alone. Future studies should incorporate additional datasets, including redshift-drift measurements, standard sirens, and high-redshift galaxy observations, to refine the estimate of A and test the model’s predictions more rigorously.

In summary, TEEM offers a promising foundation for an energy-based cosmology, but further theoretical development and multi-probe observational testing are required to fully establish its validity.

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